

Reproduce-then-Extend — Civil-War Onset (Fearon & Laitin 2003)

(Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Fearon, J. D. & Laitin, D. D. (2003). “Ethnicity, Insurgency, and Civil War.” *American Political Science Review* 97(1):75–90. A **logistic regression** of civil-war onset on 11 country-year predictors (Sambanis 2006 imputed data; Harvard Dataverse doi: 10.7910/DVN/KRKWK8). Onset is a **rare event** (~1.6%); this example illustrates **out-of-sample classification** and the right evaluation metric for imbalanced data.

1 Background

Fearon and Laitin ask **what conditions make civil war likely** worldwide from 1945 to 1999. Their central aim is to adjudicate between two explanations: civil war as a product of ethnic and religious **grievance**, versus civil war as a product of conditions that favor **insurgency** – weak states, poverty, rough terrain, and populations governments cannot control. They use **logistic regression** of civil-war onset on structural covariates to test which set of factors predicts onset. The **primary conclusion** is that insurgency-favoring conditions (low GDP per capita, large population, mountainous terrain, political instability, new states) strongly predict civil war, while ethnic and religious fractionalization do **not** – a direct challenge to the ethnic-grievance thesis.

2 Data and codebook

Unit of analysis: country-year; n = 7,140 (116 civil-war onsets, ~1.6%).

Variable	Definition
warstds	civil-war onset: 1 in the first year of a new civil war (outcome)
warhist	prior civil-war history in the country
ln_gdpen	log GDP per capita (lagged one year)
lpopns	log national population
lmtnest	log % of the country that is mountainous terrain
ncontig	1 if the state has noncontiguous territory
oil	1 if fuel exports exceed one-third of total exports (oil exporter)
nwstate	1 in the first two years of a newly independent state
inst3	political instability: a ≥ 3 -point Polity change in the prior 3 years
pol4	Polity score (-10 autocracy to +10 democracy)
ef	ethnolinguistic fractionalization
relfrac	religious fractionalization

3 Descriptive results

```
d <- read.csv("data/SambnisImp.csv")
fl <- c("warhist", "ln_gdpen", "lpopns", "lmtnest", "ncontig", "oil", "nwstate",
       "inst3", "pol4", "ef", "relfrac")
d <- d[complete.cases(d[, c("warstds", fl)]), c("warstds", fl)]

round(t(sapply(d, function(x) c(mean = mean(x), sd = sd(x), min = min(x), max =
  ↪ max(x)))), 2)

##          mean    sd    min    max
## warstds  0.02 0.13   0.00   1.00
## warhist  0.17 0.38   0.00   1.00
## ln_gdpen  0.76 1.00  -3.04   4.07
## lpopns   15.75 1.63  11.01  20.95
## lmtnest   2.18 1.34   0.00   4.56
## ncontig   0.18 0.36   0.00   1.00
## oil       0.12 0.32   0.00   1.00
## nwstate   0.03 0.16   0.00   1.00
## inst3     0.15 0.34   0.00   1.00
## pol4     -0.10 7.26 -10.00  10.00
## ef        0.46 0.25   0.00   1.00
## relfrac   0.36 0.21   0.00   0.78

par(mfrow = c(1, 2))
par(mar = c(4, 4, 2.5, 1))
barplot(table(d$warstds), names.arg = c("no onset", "onset"),
        col = c("grey75", "#1f78b4"), main = "Outcome: warstds", ylab = "count")

# standardized mean difference (onset - non-onset) for each predictor
smd <- sapply(d[fl], function(x) (mean(x[d$warstds == 1]) - mean(x[d$warstds == 0])) /
  ↪ sd(x))
smd <- tail(smd[order(abs(smd))], 8)
par(mar = c(4, 6, 2.5, 1))
barplot(smd, horiz = TRUE, las = 1, cex.names = 0.85,
        col = ifelse(smd > 0, "#1f78b4", "#e31a1c"), main = "Onset vs non-onset",
        xlab = "std. mean difference")
abline(v = 0, col = "grey50")
```

4 Reproduce the published regression

```
fl_logit <- glm(warstds ~ warhist + ln_gdpen + lpopns + lmtnest + ncontig + oil +
               nwstate + inst3 + pol4 + ef + relfrac, data = d, family = binomial)
summary(fl_logit)
```

```
##
## Call:
## glm(formula = warstds ~ warhist + ln_gdpen + lpopns + lmtnest +
##      ncontig + oil + nwstate + inst3 + pol4 + ef + relfrac, family = binomial,
##      data = d)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
```

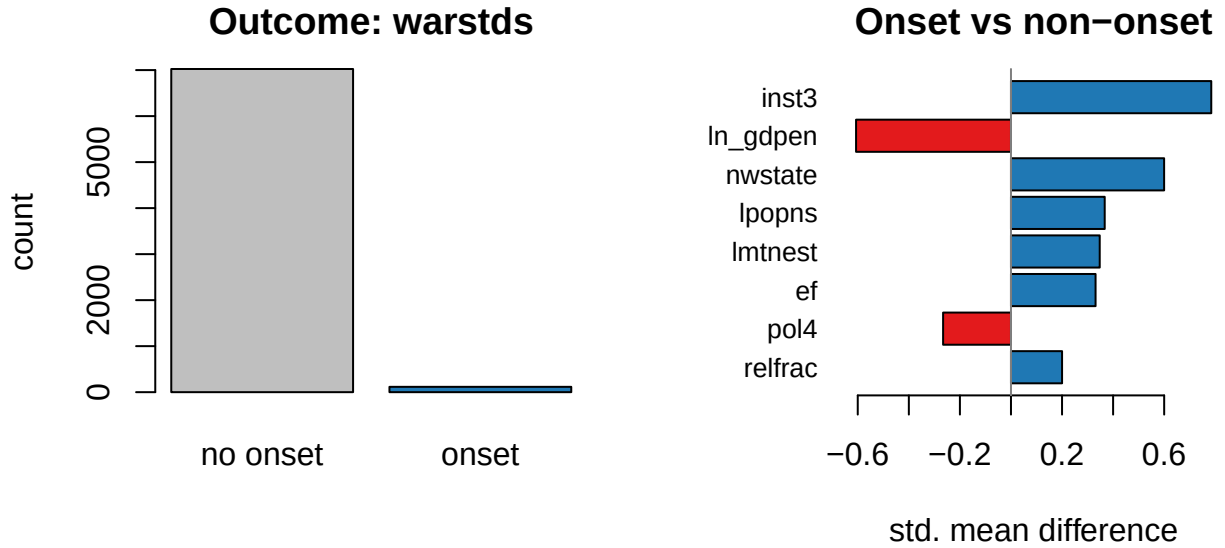


Figure 1: Class balance of the rare onset outcome (left) and the covariates that differ most between onset and non-onset years (right).

```
## (Intercept) -8.46582    1.11539   -7.590 3.20e-14 ***
## warhist      0.02247    0.25379    0.089 0.92946
## ln_gdpen     -0.35196    0.11791   -2.985 0.00284 **
## lpopns       0.18788    0.06815    2.757 0.00584 **
## lmtnest      0.18706    0.08049    2.324 0.02013 *
## ncontig      0.26477    0.28232    0.938 0.34831
## oil          0.33689    0.30155    1.117 0.26391
## nwstate      1.86032    0.32037    5.807 6.37e-09 ***
## inst3        1.36212    0.20709    6.577 4.79e-11 ***
## pol4         -0.02435    0.01781   -1.367 0.17170
## ef           0.44421    0.42340    1.049 0.29411
## relfrac      0.59413    0.50782    1.170 0.24202
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1185.9  on 7139  degrees of freedom
## Residual deviance: 1057.2  on 7128  degrees of freedom
## AIC: 1081.2
##
## Number of Fisher Scoring iterations: 7
```

Confirmation against the published study. The coefficients carry the signs and significance reported in Fearon & Laitin’s Table 1: civil war is more likely in **poorer** countries (*ln_gdpen* negative), **more populous** ones (*lpopns* positive), **more mountainous** terrain (*lmtnest* positive), **new states** (*nwstate* positive) and amid **political instability** (*inst3* positive), while ethnic/religious fractionalization are not significant — their central claim that *insurgency conditions, not ethnic grievance, drive civil war*.

5 Out-of-sample prediction with the regression

We fit on a random 70% and score the held-out 30%. With a rare outcome, accuracy is useless (predicting “no war” everywhere scores 98%); the honest evaluation is the **ROC** and especially the **precision-recall** curve.

```

set.seed(2026)
train <- sample(nrow(d), floor(0.7 * nrow(d)))
test  <- setdiff(seq_len(nrow(d)), train)

fit <- glm(warstds ~ ., data = d[train, ], family = binomial)
pred <- predict(fit, d[test, ], type = "response") # predicted onset probability
y    <- d$warstds[test]

roc <- roc.curve(scores.class0 = pred[y == 1], scores.class1 = pred[y == 0], curve =
  ↪ TRUE)
pr <- pr.curve(scores.class0 = pred[y == 1], scores.class1 = pred[y == 0], curve = TRUE)
cat(sprintf("Held-out test: %d country-years, onset base rate %.1f%%\n", length(test),
  ↪ 100 * mean(y)))

## Held-out test: 2142 country-years, onset base rate 1.6%

cat(sprintf(" Out-of-sample ROC-AUC = %.3f\n Out-of-sample PR-AUC = %.3f (a random
  ↪ model scores %.3f)\n",
  roc$auc, pr$auc.integral, mean(y)))

## Out-of-sample ROC-AUC = 0.819
## Out-of-sample PR-AUC = 0.068 (a random model scores 0.016)

```

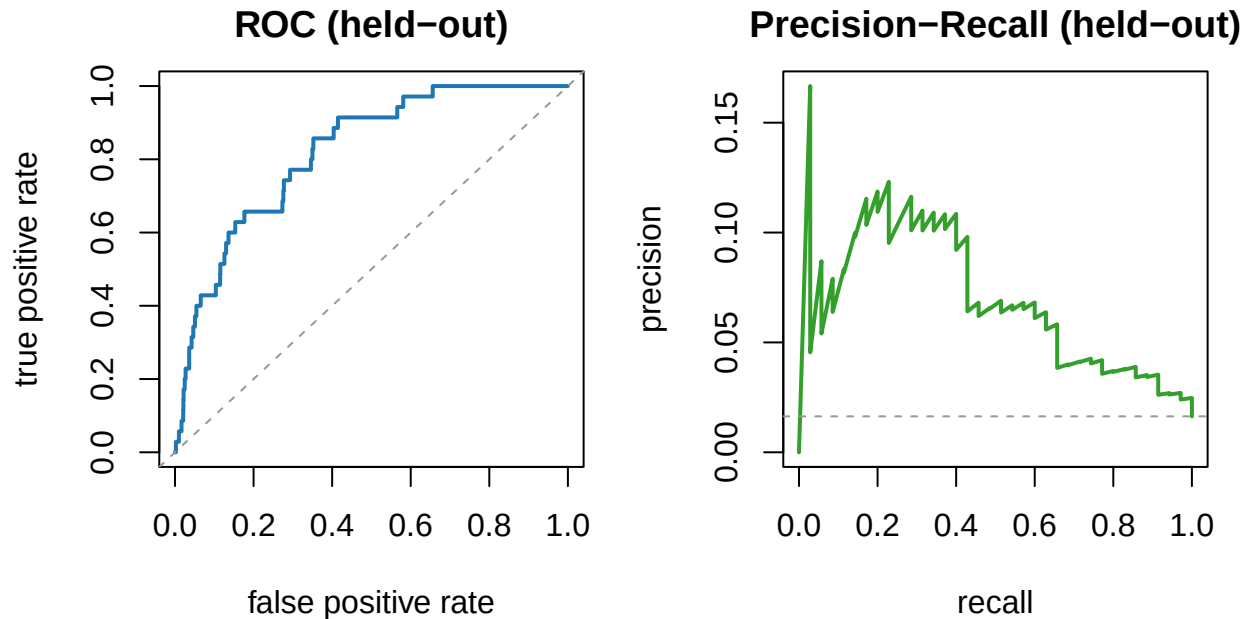


Figure 2: Out-of-sample ROC curve (left) and precision-recall curve (right) for the held-out logit predictions.

The ROC-AUC looks strong (~ 0.8), but the **PR-AUC** is far more modest — the honest picture when only 1.6% of cases are onsets. This is the key evaluation lesson: *on imbalanced data, ROC-AUC flatters a model and precision-recall tells the truth.*

6 Takeaway

This example illustrates **out-of-sample classification with a published regression** and the **evaluation-metric choice** that matters most for rare events. The Fearon–Laitin logit predicts onset above chance out

of sample, but reporting precision–recall (not accuracy or ROC alone) is what keeps the assessment honest. (A later example — Gibler–Braithwaite — extends this rare-event setting with tree-based learners.)

7 Recommended exercises

1. Replace the single split with stratified 5-fold CV and report mean AUC-PR (the outcome is ~1.6% positive, so use AUC-PR, not accuracy).
2. Fit an L1-penalized logistic regression and compare its AUC-PR and retained variables to the full logit.
3. Plot the calibration (reliability) curve of the predicted probabilities and comment on over/under-confidence.